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RAJALAKSHMI ENGINEERING COLLEGE

An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

Continuous Assessment Test-I [CAT -II]

Year : I
Semester : II
Branch : B.E CSE, EEE, AI&ML, CSE(CS)
Sub. Code : CS23231
Subject : Data Structures
QP Code : 020071

Date: 03.06.2026

Regulations 2023

Time : 120 Min

Max. Marks : 75

Answer All Questions

Part A: 10 x 2 = 20 Marks

- 1 Differentiate between Binary Search Tree (BST) and Binary Heap [CO3] [A2]
in terms of structure and usage.
- 2 Why AVL trees guarantee better performance in search operations [CO3] [A2]
compared to unbalanced BSTs.
- 3 Define the following terms in graph theory: (i) Degree of a vertex, [CO4] [A1]
(ii) Path
- 4 Why is it important to mark nodes as visited in DFS and BFS? What [CO4] [A1]
issue may arise if this is not done?
- 5 Given the graph with vertices {A, B, C, D} and directed edges [CO4] [A1]
{(A→B), (A→C), (B→D), (C→D)}, represent the graph using an
adjacency matrix.
- 6 How many topological orderings are possible for the DAG: A→B, [CO4] [A2]
A→C? List them and explain the logic.
- 7 Apply Insertion Sort to the array: [6, 2, 4, 1] and write the array [CO5] [A1]
after two iterations.
- 8 What is a hash function? State any two. [CO5] [A2]
- 9 Explain why open addressing can cause clustering problems in hash [CO5] [A1]
tables. What is primary clustering?
- 10 An array of size n is sorted using Merge Sort. What is the total [CO5] [A2]
number of comparisons in the worst case? Give the formula and
the answer for n = 8.

Part B: 5x11=55 Marks

- 11 a. Construct a Binary Heap (Min Heap) for the following set of elements in the given order: 15, 8, 10, 5, 20, 25, 3. Show all intermediate steps and the final heap structure as a binary tree and as an array. [CO3] [B1]

[OR]

- b. Discuss the drawbacks of Binary Search Trees when inserted with sorted data. How do AVL trees overcome this issue? Explain with example structures and time complexity implications. [CO3] [B2]

- 12 a. 0 1 2 3 [CO4] [B1]

0 [0 1 0 1]

1 [1 0 1 0]

2 [0 1 0 1]

3 [1 0 1 0]

Write the Algorithm and Perform BFS traversal on the same adjacency matrix starting from vertex 0 and list the visited nodes in order.

[OR]

- b. A university course scheduling system must ensure that all prerequisite courses are completed before a student takes an advanced course. Represent the course structure as a directed graph and apply Topological Sort to determine a valid order of course completion. [CO4] [B2]

Course Code	Course Name	Prerequisite(s)
C1	Introduction to Programming	—
C2	Data Structures	C1
C3	Algorithms	C2
C4	Databases	C1
C5	Operating Systems	C2
C6	Software Engineering	C3, C4

- 13 a. A power grid company is planning to connect four towns with electric cables in such a way that all towns are connected with minimum total wiring cost. The possible connections between the towns and their associated wiring costs (in lakhs of rupees) are shown below: [CO4] [B1]

- Town 1 to Town 2 $\rightarrow$ ₹1 lakh
- Town 1 to Town 3 $\rightarrow$ ₹4 lakhs
- Town 2 to Town 4 $\rightarrow$ ₹2 lakhs
- Town 3 to Town 4 $\rightarrow$ ₹3 lakhs

This setup can be represented as an undirected weighted graph with vertices $\{1, 2, 3, 4\}$.

- Model the problem as a graph.
- Apply Prim's Algorithm starting from vertex 1 (Town 1) to construct the Minimum Spanning Tree (MST).
- Write the Algorithm

[OR]

- b. Apply Dijkstra's Algorithm to find the shortest path from a source city to all other cities in the following weighted directed graph. [CO4] [B2]

Directed Edges with Weights (Road Distances):

- $A \rightarrow B$ (4)
- $A \rightarrow C$ (2)
- $B \rightarrow C$ (5)
- $B \rightarrow D$ (10)
- $C \rightarrow E$ (3)
- $E \rightarrow D$ (4)
- $D \rightarrow E$ (6)

Construct and Graph Apply Dijkstra's Algorithm starting from city A. Write the algorithm.

- 14 a. An e-commerce app needs to sort product prices quickly during a flash sale. Model the prices as an array $[55, 30, 80, 20, 60, 75, 90]$. Apply Quick Sort using the first element as pivot and demonstrate the recursive partitioning process. Write algorithm. [CO5] [B1]

[OR]

- b. A hospital wants to store patient IDs efficiently in a hash table of size 10 using the division method. Insert the keys [23, 43, 13, 27, 33, 42, 88] using open addressing with linear probing. Show the final hash table and how collisions are resolved. [CO5] [B2]

- 15 a. An online exam portal uses hashing to store student login data. Collisions are frequent due to similar IDs. Compare open addressing and separate chaining with example and discuss which one would be more efficient for this use case. [CO5] [B1]

[OR]

- b. A shopping cart system observes performance drops as more items are added. Hash collisions increase in the table of size 5. Explain when and how rehashing should be done. Simulate rehashing with the keys [10, 20, 15, 25, 30] to a new table of size 10. [CO5] [B2]

Reg. No.

2 1 1 6 2 3 1 0 0 1 2 5 5



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Continuous Assessment Test-II [CAT-II]

Year : I
Semester : II
Branch : B.E./B.Tech.
Sub. Code : CS23231
Subject Name : Data Structures
QP Code : 071002

[Regulations 2023]

[Common to CSE, IT, AI&ML, CSD, AI&DS and CSE (Cyber Security)]

Date: 12.06.2024

Time: 120 Minutes

Marks: 75

Answer ALL Questions

Part A [2 x 2 = 4 Marks]

- 3.4. What is the purpose of a binary heap in priority queue implementation? [A2] CO3
- 3.5. Name two properties of AVL trees that ensure balance. [A1] CO3

Part B [1 x 11 = 11 Marks]

- 3.6. a. Explain the procedure for inserting a new element in a binary heap and discuss its time complexity. [B1] CO3

[OR]

- b. Given the following sequence of numbers: [10, 20, 30, 25, 15, 5, 35], construct an AVL tree. Show each step of insertion and the corresponding rotations required to maintain balance. [B2] CO3

Answer ALL Questions

Part A [4 x 2 = 8 Marks]

- 4.1. How is an adjacency matrix used to represent a graph? [A1] CO4
- 4.2. Compare the efficiency of Depth First Search (DFS) and Breadth First Search in terms of Space Complexity [A2] CO4
- 4.3. Define the term "minimum spanning tree" and state its significance in network design. [A1] CO4
- 4.4. What is the main difference between Prim's algorithm and Kruskal's algorithm? [A1] CO4

Part B [2 x 11 = 22 Marks]

- 4.5. a. In the following graph, find the shortest path from node A to all other nodes using Dijkstra's algorithm. Show each step of the process. [B2] CO4
- A - 1 - B
A - 4 - C
B - 2 - C
B - 5 - D
C - 1 - D

- b. Given a network of computers, use Prim's algorithm to find the minimum spanning tree for the following graph. Show each step and the resulting MST. [B1] CO4

A - 2 - B

A - 3 - C

B - 1 - D

C - 4 - D

C - 5 - E

- 4.6. a. Use Depth First Search (DFS) to detect a cycle in the following graph. Explain each step. [B2] CO4

A → B

B → C

C → D

D → B

- b. Discuss the implementation of Breath First Search (BFS) using both recursive and iterative approaches. [B2] CO4

[OR]

Answer ALL Questions

Part A [4 x 2 = 8 Marks]

- 5.1. What is the main advantage of using insertion sort for nearly sorted data? [A1] CO5
 5.2. Define quick sort and its pivot selection strategy. [A2] CO5
 5.3. What is a hash function? [A1] CO5
 5.4. Name two collision resolution techniques used in hashing. [A1] CO5

Part B [2 x 11 = 22 Marks]

- 5.5. a. Use merge sort to sort the array [50, 20, 40, 10, 30, 60]. Show each step of the process and the final sorted array. [B2] CO5

[OR]

- b. Sort the following list using quick sort: [35, 10, 55, 20, 5, 25]. Show each step and the final sorted list. [B1] CO5

- 5.6. a. A hash table uses separate chaining to resolve collisions. Insert the following keys into the hash table with a size of 7 using the hash function $h(x) = x \% 7$: [10, 20, 5, 15, 25]. Show the final table. [B1] CO5

[OR]

- b. Resolve collisions using open addressing with quadratic probing for the following keys into a hash table of size 7: [10, 20, 30, 40]. Use the hash function $h(x) = x \% 3$ and show each step. [B2] CO5



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Sub. Code : CS23231
Subject : Data Structures
QP Code : 021071

Date: 03.06.2025

(Regulations 2023)

Time : 120 Min

Max. Marks : 75

Answer All Questions

Part A: 10 x 2 = 20 Marks

1. Define an AVL tree. What property does it maintain after every insertion or deletion? [CO3] [A1]
2. A sequence of numbers 1, 2, 3, ..., n is inserted into an initially empty Binary Search Tree in ascending order. What will be the height of the resulting tree? [CO3] [A2]
3. Represent the following undirected graph using an adjacency list: vertices = {A, B, C, D}, edges = {(A,B), (A,C), (B,D)}. [CO4] [A1]
4. What is an adjacency matrix and how is it used to represent a graph? [CO4] [A1]
5. A telecom company wants to connect multiple cities with the minimum cable cost. Which graph algorithm is suitable and why? [CO4] [A2]
6. Why can't we apply topological sort on a graph with cycles? Justify with an example. [CO4] [A2]
7. If an array is already sorted, what is the total number of comparisons made in Insertion Sort? [CO5] [A2]
8. Use the division method to hash the keys 24, 56, 39 into a table of size 10. Show the resulting indices. [CO5] [A2]
9. Perform a single partitioning step of Quick Sort on the array [8, 4, 7, 2, 9] using 8 as the pivot. Show the resulting left and right sub-arrays. [CO5] [A1]
10. What is rehashing? When and why is it used in a hash table? [CO5] [A2]

Part B: 5x11=55 Marks

- 11 a. Insert the sequence of numbers 50, 30, 70, 20, 40, 60, 80 [CO3] [B1]
into an empty AVL tree. After each insertion, show the tree
structure and indicate where rotations are needed to
maintain balance.

[OR]

- b. Write the pseudocode to construct a BST from a given list [CO3] [B2]
of integers (e.g., [50, 30, 70, 20, 40, 60, 80]).

- 12 a. In a college campus, different departments are connected [CO4] [B1]
through internal walkways. The navigation team wants to
model these connections as a graph so that they can build
a campus navigation app.

There are 5 departments: Computer Science
(CSE), Electrical (EEE), Mechanical (MECH), Civil
(CIVIL) and Information Technology (IT).

The walkways between departments are bidirectional and
are as follows:

- CSE is connected to EEE and MECH
 - EEE is connected to CIVIL
 - MECH is connected to CIVIL and IT
 - CIVIL is connected to IT
- i. Represent the department network as an undirected graph.
 - ii. Create the adjacency list for the graph.
 - iii. Create the adjacency matrix for the graph using the department names as indices.
 - iv. Briefly explain the advantages and disadvantages of each representation for this scenario.

[OR]

- b. A security robot is deployed in a multi-room building to [CO4] [B2]
inspect all areas for safety. The building is represented as
a set of connected rooms, where each room is a node and
each doorway is an edge between two rooms. The robot
starts in Room A and follows a Depth-First Traversal
(DFS) strategy to explore the building.

The connections between the rooms are as follows
(bidirectional):

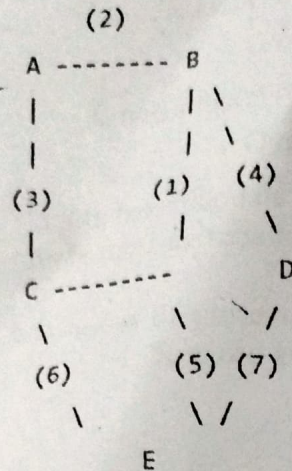
- Room A is connected to Room B and Room C
 - Room B is connected to Room D
 - Room C is connected to Room E and Room F
 - Room E is connected to Room G
- i. Represent the room layout as an undirected graph.
 - ii. Perform Depth-First Traversal starting from Room A.
 - iii. Explain how DFS helps in covering all rooms and how it differs from BFS in this use case.
 - iv. Write the algorithm.

- 13 a. A delivery company wants to optimize its routes between distribution centers in a city. The city map is modeled as a weighted directed graph. The map contains the following connections: [CO4] [B1]

- $A \rightarrow B$ (6 minutes)
 - $A \rightarrow D$ (1 minute)
 - $B \rightarrow C$ (5 minutes)
 - $D \rightarrow B$ (2 minutes)
 - $D \rightarrow E$ (1 minute)
 - $E \rightarrow C$ (5 minutes)
 - $C \rightarrow F$ (2 minutes)
 - $E \rightarrow F$ (7 minutes)
- i. Draw the directed weighted graph based on the data above.
 - ii. Apply Dijkstra's Algorithm starting from Center A.
 - iii. Show the step-by-step table for updating the shortest delivery times and the visited set.

[OR]

- b. Define Minimum Spanning Tree. Apply Prim's Algorithm to find the Minimum Spanning Tree (MST) for the following undirected weighted graph. Start the algorithm from vertex A. Show the step-by-step process of selecting edges, updating visited nodes, and calculating the total cost of the MST. Write the algorithm. [CO4] [B2]



- 14 a. A teacher wants to organize student test scores in ascending order as they are entered one by one. Model this as an Insertion Sort process. Given the scores [70, 85, 60, 90, 75], apply Insertion Sort and show the steps after each iteration. Comment on the best and worst cases of this method. [CO5] [B1]

[OR]

- b. A library system uses the mid-square method and folding method to hash book IDs. Given the keys [1234, 4321, 5678], show how these methods generate hash indices in a table of size 100. Explain how each method minimizes collisions. give the question more clearly and explain the answer [CO5] [B2]

- 15 a. A university wants to build a student portal that stores roll numbers efficiently and handles duplicate hash values using separate chaining. Hash the keys [101, 121, 131, 141, 151] into a table of size 10. Show the final structure of each index and explain how retrieval would work. [CO5] [B1]

[OR]

- b. An airline ticketing system stores ticket IDs and needs to both sort ticket costs quickly and store ticket IDs for fast retrieval. Propose a solution that uses a combination of Merge Sort and Hashing. Explain why this hybrid approach improves efficiency. [CO5] [B2]